Interpretable EEG Eye-State Detection: Comparing Feature Representations under Temporally Rigorous Cross-Validation

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Abstract

Electroencephalography (EEG) measures electrical activity of the brain through scalp electrodes, enabling non-invasive monitoring of neural states. EEG eye-state classification is the task of detecting whether a subject's eyes are open or closed from scalp electrode recordings. This paper addresses limitations of current EEG datasets, such as temporal data leakage, SNR constraints of consumer-grade hardware, and the resulting limitations on spectral feature extraction. It used the UCI EEG Eye State dataset, which was recorded with an Emotiv Epoc headset. I preprocessed the data in sliding windows, performed per-window mean subtraction, artifact rejection, blocked cross-validation, and performed feature extraction of spectral, statistical, and raw EEG data on Random Forest, LSTM, and transformer models. The LSTM outperformed all Random Forest models with 0.68 ROC-AUC, and a subsequent transformer experiment achieved 0.81 ROC-AUC using RoPE positional encoding and blocked cross-validation. These results suggest that temporal modeling outperforms hand-crafted spectral features for consumer-grade EEG classification, and that methodological rigor in train-test splitting is critical for reliable performance estimation.

1. Introduction

EEG-based eye-state classification has practical applications in drowsiness detection for driver safety monitoring and in the development of brain-computer interfaces that enable hands-free control for users with motor impairments.

The alpha rhythm is the relaxed state of eye closure, characterized by 8-13 Hz oscillations originating in occipital regions, and it disappears when eyes open because the visual cortex shifts from a resting to active state. This makes EEG eye-state classification theoretically tractable because the alpha rhythm provides a reliable neural correlate of eye state that should be detectable from scalp electrodes.

Most existing work uses random train/test splits that inflate performance through data leakage. Consumer-grade hardware introduces noise that challenges spectral methods (Nolan et al., 2010). The UCI EEG Eye State dataset, recorded using a consumer-grade Emotiv Epoc headset, provides a standard benchmark for evaluating these challenges.

This paper presents an evaluation of multiple feature representations and classification architectures for EEG eye-state detection using the UCI EEG Eye State dataset. Random Forest models of different features were compared to an LSTM on raw EEG, with a transformer experiment extending the comparison. Results demonstrated that the LSTM outperformed all Random Forest models, and a subsequent transformer experiment further improved performance, achieving 0.81 ROC-AUC. These findings suggest that attention over raw temporal windows captures discriminative structure that handcrafted features may miss.

2. Methods

2.1 Dataset and Preprocessing

The dataset was a 117-second continuous recording of EEG eye states. There were 14980 samples (meaning around 128 samples per second) and 14 channels. Each label is either a "0," indicating eyes open, or a "1," indicating eyes closed. Data was preprocessed in sliding windows of size 128 and stride 32. Per-window mean subtraction was implemented to remove DC drift. Artifact rejection was then implemented; windows where any channel exceeded 150μV peak amplitude were discarded. Train and test sets were split through blocked cross-validation, not random split, to prevent the model from training on data that is temporally adjacent to test data, which would inflate performance due to the non-stationary nature of the EEG recording. A 4-window boundary gap was utilized to prevent overlap leakage between training and test windows. Data was divided into 5 contiguous folds of equal size, with each fold serving as the test set once while the remaining folds formed the training set. StandardScaler was fitted on training data only per fold.

2.2 Feature Extraction

A real-valued Fast Fourier Transform (rfft) was applied to each window along the time axis to obtain the power spectrum, which was then computed by taking the absolute value of the transformation squared. The four bands were then extracted with their Hz ranges: delta ranged from 1-4 Hz, theta ranged from 4-8 Hz, alpha ranged from 8-13 Hz, and beta ranged from 13-31 Hz (consistent with standard EEG band definitions). The power within each band was averaged per channel, log compression was applied, and the output dimensionality led to 56 features concatenated across all bands. Statistical features of mean, standard deviation, minimum, maximum, skewness, and kurtosis were computed per channel along the time axis. There were 84 features concatenated across the six descriptors. Additionally, alpha increases when eyes close, while beta decreases. The ratio of alpha/beta should therefore rise during eye closure, providing a physiologically motivated feature for eye-state classification. Log compression was applied to the ratio to reduce the influence of extreme values, leading to an output dimensionality of 14 features. FFT and statistical features were then concatenated, with an output dimensionality of 140 features. Each feature representation was evaluated independently and in combination to assess the contribution of spectral versus time-domain information to classification performance.

2.3 Models

A Random Forest model of 100 estimators and random state 42 was used. It received inputs of flattened feature vectors. There was no temporal information used, as each window was treated independently. An LSTM with a single layer, hidden size of 32, and dropout of 0.3 was trained on raw EEG windows of shape (batch, 128, 14) using BCEWithLogitsLoss, the Adam optimizer with learning rate 1e-3, and gradient clipping at max_norm = 1.0. It trained for 30 epochs, determined empirically to minimize overfitting. The LSTM is architecturally suited for EEG because it processes sequential data and maintains temporal context across timesteps, unlike Random Forest which treats each window as an orderless feature vector. A 3-layer transformer with 64-dimensional embeddings was trained on raw EEG windows of shape (batch, 128, 14) using an AdamW optimizer with weight decay 1e-2 and a learning rate of 1e-4. The model also utilized RoPE positional encoding and patience-based early stopping, being trained with 5-fold blocked cross-validation. However, the ROC-AUC was computed using the first 4 folds, as fold 5 was excluded from the mean due to severe class imbalance in the test split (77 class-0 vs. 11 class-1 windows).

3. Results

3.1 Model Comparison

Model	Features	ROC-AUC	Std
Random Forest	FFT Bandpower	0.52	0.065
Random Forest	Alpha/Beta Ratio	0.46	0.058
Random Forest	Combined	0.59	0.039
Random Forest	Statistical	0.62	0.028
LSTM	Raw EEG	0.68	0.082
Transformer	Raw EEG (windowed)	0.81	0.068

Table 1: Model comparison by ROC-AUC

Table 1 presents the ROC-AUC scores and standard deviations of each model. Random Forest scored a 0.52 ± 0.065 ROC-AUC with FFT, a 0.46 ± 0.058 ROC-AUC with alpha/beta ratio, a 0.59 ± 0.039 with both spectral and statistical features, and a 0.62 ± 0.028 ROC-AUC with statistical features alone. The LSTM scored a 0.68 ± 0.082 ROC-AUC. Among Random Forest models, alpha/beta performed the worst, followed by FFT, combined spectral/statistical, and statistical alone performing the best. The LSTM performed better than the best Random Forest model (statistical). A subsequent transformer experiment achieved 0.81 ± 0.068 ROC-AUC, outperforming all other models.

3.2 Feature Interpretability Analysis

SHAP analysis was applied to the best-performing Random Forest model to identify which features most strongly influenced eye-state predictions. SHAP showed the max and kurtosis of frontal and temporal electrodes dominating since open eyes move around constantly and generate electrical artifacts in those places, which is surprising because occipital alpha was expected to drive classification based on neuroscience theory but failed to appear in the top features. This was due to the Emotiv Epoc's known poor signal quality on occipital electrodes due to hair interference. The plot shows that high kurtosis at frontal electrodes predicts eyes-open, while low kurtosis predicts eyes-closed, which tells us that eye movements during the eyes-open state generate spike-like artifacts at frontal electrodes, producing high kurtosis values. This suggests that the discriminative signal in this dataset derives primarily from eye movement artifacts rather than alpha rhythm suppression, explaining why kurtosis and maximum amplitude outperformed spectral band power features.

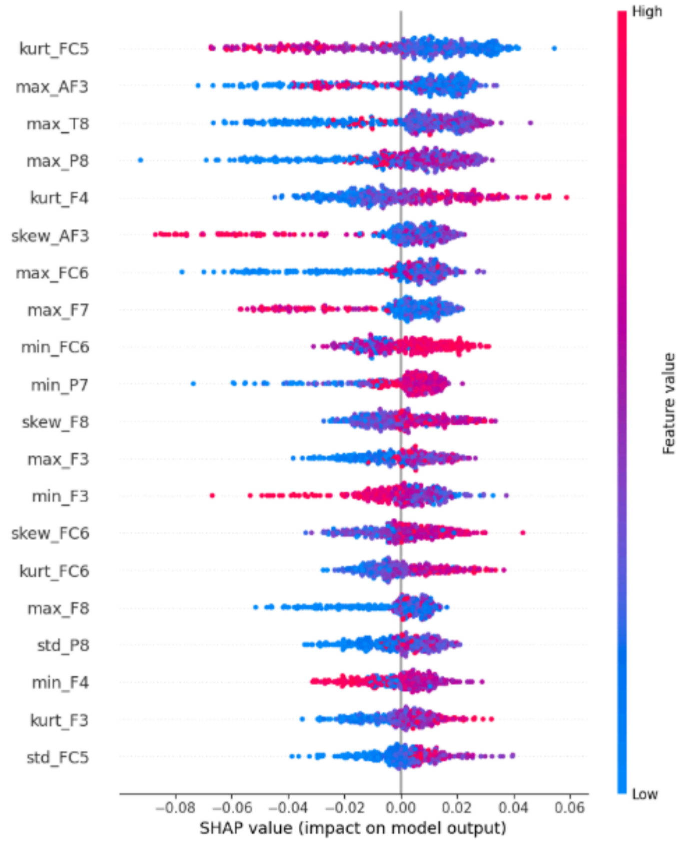

Figure 1: SHAP summary plot showing feature importance for the Random Forest statistical model. Features are ranked by mean absolute SHAP value. Pink indicates high feature values, blue indicates low feature values.

4. Discussion

Spectral features did not perform well because the alpha power difference between eyes-open and eyes-closed was only 10-19% across channels, insufficient for reliable classification given the SNR limitations of the Emotiv Epoc consumer-grade headset. Statistical features outperformed them because they capture global signal dynamics (amplitude variability, distributional shape, tail behavior) that reflect the overall change in neural activity during eye closure, without depending on precise frequency resolution that consumer hardware cannot provide. LSTM outperformed Random Forest because it processed each timestep sequentially, maintaining a hidden state that captured oscillatory patterns across the full 128-sample window. However, Random Forest treated the feature vector as orderless and lost all within-window temporal relationships. SHAP analysis further revealed that discriminative features derived primarily from eye movement artifacts at frontal electrodes rather than occipital alpha rhythm, suggesting that the dataset's classification signal is dominated by oculomotor artifacts rather than neural oscillations. The limitations of the dataset included a single session of a single subject with consumer-grade hardware, as well as a small sample size after artifact rejection. There were 441 windows limiting statistical power, and non-stationary class distributions in the final folds skewed evaluation. Future work could address these limitations by collecting multi-subject recordings with clinical-grade EEG hardware, which would provide sufficient data for deeper architectures and more reliable spectral feature extraction.

A 3-layer transformer architecture trained on raw EEG windows using 5-fold blocked cross-validation, RoPE positional encoding, and patience-based early stopping was implemented on the dataset as a follow-up experiment. It achieved an ROC-AUC of 0.81, outperforming the LSTM (0.68) and Random Forest (0.62) models, suggesting that attention over raw temporal windows captures discriminative structure that handcrafted features may miss.

5. Conclusion

This paper compared and evaluated multiple classification models for EEG eye-state detection using the UCI EEG Eye State dataset. The LSTM outperformed all Random Forest models, while statistical features were the best-performing Random Forest. This meant the SNR limitations of consumer-grade hardware and single-subject recording constrained overall classification performance, indicating future work to aim towards multi-subject recordings and clinical-grade EEG hardware to provide sufficient data. Subsequently, a transformer extension implementing RoPE positional encoding achieved 0.81 ROC-AUC, outperforming the LSTM baseline and suggesting that attention-based architectures warrant further investigation on higher-quality EEG datasets.

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